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Motor for washing machine

Abstract

The motor for a washing machine includes a motor housing having a stator, a pair of first supports formed integrally with the motor housing in the front of the motor housing, a connection cylindrical member cylindrically extending backwardly from a middle part of the motor housing to be formed integrally with the motor housing, and a pair of second supports integrally formed at right and left ends of the connection cylindrical member.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a motor for a washing machine. More specifically, the present invention relates to a motor for a washing machine capable of shortening a motor assembling process and reducing the overall size of the motor to easily secure a space for installing the motor, by forming a motor housing in which a stator and a rotor are placed integrally with a fixing bracket

by injection molding simultaneously.

BACKGROUND ART

(2) In general, a washing machine includes a rotating drum of a cylindrical shape which contains laundry and a motor which is connected to a rotating shaft in the rear of the rotating drum with a belt to drive the rotating drum. The motor is fixed to the main body of a washing machine or drying machine. A housing or bracket for fixing the motor is made of aluminum or a bulk molding compound (BMC) resin.

(3) In the case of using aluminum as in Korean Patent No. 10-1185121, noise or vibration occurs at the site of fixing the motor, and the material cost is high. Thus, recently a BMC resin has been used for motor housings and brackets. In the case of using a BMC resin, noise or vibration remarkably decreases, the material cost is saved, and the product is resistant to moisture.

(4) A motor using a BMC resin is configured to stably drive a drum with a belt while being supported by a fixing bracket, by installing the fixing bracket inside the main body of the washing machine after the fixing bracket and the motor housing which are made of a BMC resin have been fastened to each other. This structure is disclosed in Korean Patent No. 10-1707142. According to said prior art reference, a motor is fixed to legs for supporting the motor by forming coupling holes through the lower surface of the motor housing, and fastening coupling bolts passing through fixing holes in the first and second legs to the coupling holes.

(5) However, when the coupling bolts are fastened to the coupling holes having screw threads formed through the lower surface of the motor housing, cracks occur inside the coupling holes because the motor housing is made of a BMC material, and the fastening force decreases, disallowing the motor from retaining stable coupling. In addition, durability significantly decreases.

(6) Further, the legs which are at least one or more are each coupled to the motor housing, which requires a complicate assembling process. Moreover, a reinforcement structure is not provided between the first and second legs, disallowing the motor from being stably coupled to the main body of a drying machine. In addition, the coupling bolts are screw-assembled while the fixing holes of the first and second legs match the coupling holes of the motor housing, and thus productivity is reduced because it is difficult to exactly match the fixing holes to the coupling holes.

(7) Accordingly, the present inventors suggest a motor for a washing machine having a novel structure, in order to solve the above problems.

SUMMARY OF INVENTION

Technical Task

(8) It is an object of the present invention to provide a motor for a washing machine of a novel structure which couples the motor stably and shortening a process for assembling a motor housing and a fixing bracket.

(9) It is another object of the present invention to provide a motor for a washing machine of a novel structure providing enhanced reinforcement to a fixing bracket formed integrally with a motor housing.

(10) The objects above and other objects inferred therein can be easily achieved by the present invention explained below.

Means for Solving Technical Task

(11) The motor for a washing machine according to the present invention comprises a motor housing **11** comprising a stator **12**; a pair of first supports **16** formed integrally with the motor housing **11** in the front of the motor housing **11**; a connection cylindrical member **11**B cylindrically extending backwardly from a middle part of the motor housing **11** to be formed integrally with the motor housing **11**; and a pair of second supports **21** integrally formed at right and left ends of the connection cylindrical member **11**B.

(12) In the present invention, a curved guide **23** having a structure of integrally connecting upper

parts of the second supports **21** may be formed integrally with the connection cylindrical member **11**B.
(13) In the present invention, a support reinforcer **24** having a plurality of hollows may be integrally formed downwardly of the curved guide **23**.
(14) In the present invention, a movement preventing support pad **25** being in surface-contact with a bottom surface of a main body of a washing machine may be integrally formed in a lower part of the support reinforcer **24**.
(15) In the present invention, a nonslip embossed part **25**A may be formed in a lower surface of the movement preventing support pad **25**.
(16) In the present invention, a rear bearing cover **19** may be coupled to an insertion space **11**B′ in the connection cylindrical member **11**B.
Effect of the Invention
(17) The motor for a washing machine of the present invention has a novel structure forming a fixing bracket integrally with a motor housing in the front, capable of shortening a motor assembling process to increase productivity, thereby saving the cost incurred to manufacture the motor, and also providing a reinforcing force to the fixing bracket, thereby allowing the fixing bracket to be firmly fixed and supported inside a main body of a washing machine to prevent noise and vibration.
(18) In addition, the present invention forms the rear fixing bracket integrally with the front motor housing at the back end thereof while a stator and a rotor are formed in the motor housing by insert injection molding, thereby making a compact motor and minimizing a space for installing the motor.

Description

BRIEF OF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a front perspective view of a motor for a washing machine according to the present invention;
(2) FIG. **2** is a rear perspective view of the motor for a washing machine according to the present invention;
(3) FIG. **3** is a front exploded perspective view of the motor for a washing machine according to the present invention;
(4) FIG. **4** is a rear exploded perspective view of the motor for a washing machine according to the present invention;
(5) FIG. **5** is a cross-sectional view of the motor for a washing machine according to the present invention; and
(6) FIG. **6** is a rear perspective view of a motor for a washing machine according to another embodiment of the present invention.
(7) Hereinafter, the present invention will be described in detail with reference to the accompanying drawings.
DETAILED DESCRIPTION FOR CARRYING OUT THE INVENTION
(8) FIG. **1** is a front perspective view of a motor **10** for a washing machine according to the present invention. FIG. **2** is a rear perspective view of the motor **10** for a washing machine according to the present invention. FIG. **3** is a front exploded perspective view of the motor **10** for a washing machine according to the present invention. FIG. **4** is a rear exploded perspective view of the motor **10** for a washing machine according to the present invention. FIG. **5** is a cross-sectional view of the motor **10** for a washing machine according to the present invention.
(9) The motor **10** according to the present invention is configured to stably rotate a rotating drum of a washing machine or a drying machine with a belt, etc., while the motor **10** is supported by a

fixing bracket **20**, by installing the fixing bracket **20** formed integrally with a motor housing **11** inside a main body (not shown) of a washing machine. As used herein, “a motor for a washing machine” is mainly described; however, the present invention is not necessarily limited to a motor for a washing machine but may be applied to a motor for a drying machine or a motor for other uses.

(10) As illustrated in FIGS. **1** and **2**, the motor **10** for a washing machine according to the present invention is provided by forming the motor housing **11** integrally with the fixing bracket **20** by injection molding.

(11) The motor **10** according to the present invention is assembled without a process of assembling the fixing bracket **20**, by simultaneously forming, by injection molding, the motor housing **11** which is made by disposing a stator **12** in an insert injection mold to be formed of a bulk molding compound (BMC) resin integrally with the fixing bracket **20** in the rear of the motor housing **11**.

(12) A power connector insert **14** installed at one side of the circumference in the front of the motor housing **11** comprising the stator **12** may be formed integrally with the motor housing **11** by injection molding.

(13) Through a side slide mold between an upper mold and a lower mold, the front part of the motor housing **11** formed integrally with the fixing bracket **20** is disposed in the upper mold (not shown), the middle part of the motor housing **11** is disposed in the side slide mold (not shown), and the part of the fixing bracket **20** is disposed in the lower mold (not shown), to be integrally formed by injection molding at the same time.

(14) As used herein, the “front part” of the motor housing **11** refers to the side of a power connector insert **14** and a pair of first supports **16** having a support cylindrical body **16**A at the lower end, which are inclined downwardly from both sides of a front reinforcing cylindrical body **11**A through which a rotating shaft **15** passes. The front part is disposed in the upper mold. A coupling hole **16**A′ is formed in the center of the support cylindrical body **16**A. A power connector inserting hole **17** is formed inside the power connector insert **14**.

(15) As used herein, the “middle part” of the motor housing **11** refers to a part where the stator **12** is located. The middle part is formed by insert injection molding while being disposed in the side slide mold, and then the side slide mold is opened. Thereafter, the upper and lower molds are opened simultaneously, such that the fixing bracket **20** can be formed integrally with the motor housing **11** in the rear of the “middle part” of the motor housing **11** by injection molding at the same time.

(16) A rotor **13** coupled to the rotating shaft **15** is inserted frontwardly from the rear of the stator **12** located in the motor housing **11** while the motor housing **11** comprising the stator **12** is formed by injection molding, and a drive pulley **15**A is coupled to the front end of the rotating shaft **15** passing through the center of the reinforcing cylindrical body **11**A. The drive pulley **15**A is coupled to the front of the rotating shaft **15** and rotate together with the rotating shaft **15** to drive a drum (not shown) of a drying machine or a washing machine with a belt (not shown).

(17) A front bearing **15**B supports the rotation of the rotating shaft **15** in the front side of the rotating shaft **15**. The front bearing **15**B is coupled to the inside of the reinforcing cylindrical body **11**A. FIGS. **3** and **4** illustrate the stator **12** separated from the motor housing **11**, only for convenience in explanation. The stator **12** is not separated from the motor housing **11** in practice, because the motor housing **11** is formed by disposing the stator **12** in an insert injection mold.

(18) A rear bearing **18** is coupled to the rear end of the rotating shaft **15**, and the rear bearing **18** is coupled to a rear bearing cover **19**. A connection cylindrical member **11**B cylindrically extending backwardly from the middle part of the motor housing **11** is formed integrally with the motor housing **11**. An insertion space **11**B′ for coupling the rear bearing cover **19** is formed in the connection cylindrical member **11**B. The rotor **13** is inserted into the motor housing **11** through the insertion space **11**B′.

(19) A pair of second supports **21** formed integrally with the connection cylindrical member **11**B at

right and left lower ends are formed integrally with the motor housing **11**. A curved guide **23** for assembling the rotating shaft and the rear bearing cover may be integrally formed in the center between one second support **21** and another second support **21**. The front of the curved guide **23** may be formed integrally with the lower end of the connection cylindrical member **11**B. A support cylindrical body **21**A is formed at the lower end of the second support **21**.

(20) The rotor **13** coupled with the rotating shaft **15** may be inserted into the insertion space **11**B′ of the connection cylindrical member **11**B while being guided through the curved guide **23**, facilitating accurate and smooth assembling of the rotor **13** and the rear bearing cover **19** having a cylindrical shape. Further, when the rear bearing **18** in the rear bearing cover **19** is worn out, the bearing can be easily replaced and repaired.

(21) A support reinforcer **24** having a plurality of hollows is integrally formed downwardly of the rotating shaft **15** and the curved guide **23**, and both lower ends of the support reinforce **24** are connected and formed integrally with the support cylindrical bodies **21**A of the pair of second supports **21**. Accordingly, the second supports **21** are firmly supported, so as to not only withstand the load of the motor housing **11** which is a heavy weight containing the stator **12** and the rotor **13**, but also be supported in balance with the first supports **16** in the front. Thus, the motor housing **11** does not incline in one direction, and the motor **10** can be installed stably.

(22) The present invention forms the motor housing and the fixing bracket integrally by molding, using the connection cylindrical member **11**B therebetween. Accordingly, the fixing bracket **20** may be formed in a short length without extending backwardly, capable of minimizing the size of the motor **10**. In addition, the curved guide **23** may be open upwardly, which facilitates the inserting and disassembling of the rear bearing cover **19**.

(23) FIG. **6** is a rear perspective view of a motor **10** for a washing machine according to another embodiment of the present invention. As illustrated in FIG. **6**, the present invention inhibits back and forth and right and left movement of the motor **10** caused by vibration occurring upon operation of the motor **10**, thereby preventing loosening of a coupling bolt (not shown) which is inserted into a coupling hole **16**A′ in a support cylindrical body **16**A of a first support **16** and a coupling hole **21**A′ in a support cylindrical body **21**A of a second support **21** and is screw-coupled to a main body of a washing machine, to respond to vibration of the motor and retain firm installation.

(24) A movement preventing support pad **25** that is in surface-contact with the bottom surface of the main body of a washing machine is integrally formed in the vertical lower part of a support reinforce **24** below a curved guide **23**, such that the lower surface of the support cylindrical body **21**A of the second support **21** and the lower surface of the movement preventing support pad **25** are in surface-contact with the bottom surface of the main body of a washing machine. Accordingly, a force to move the motor **10** back and forth or right and left which may be generated when vibration occurs upon operation of the motor **10**, may be inhibited. Thus, the motor **10** can be retained in more stable and firm installation.

(25) In addition, nonslip embossed parts **21**A″, **25**A are formed in the lower surface of the support cylindrical body **21**A of the second support **21** and the lower surface of the movement preventing support pad **25**, respectively, to prevent back and forth or right and left movement of the support cylindrical body **21**A of the second support **21** and the movement preventing support pad **25** upon vibration of the motor **10**.

(26) The nonslip embossed part **21**A″ and the nonslip embossed part **25**A may have embossings formed in the vertical or horizontal direction. Alternatively, the nonslip embossed part **21**A″ and the nonslip embossed **25**A may have embossings formed in opposite directions.

(27) The detailed description of the present invention described as above simply explains examples for understanding the present invention, but does not intend to limit the scope of the present invention. The scope of the present invention is defined by the accompanying claims. Additionally,

it should be construed that simple modifications or changes of the present invention fall within the scope of the present invention.

Claims

1. A motor for a washing machine, comprising: a motor housing (**11**) comprising a stator (**12**); a pair of first supports (**16**) formed integrally with the motor housing (**11**) in the front of the motor housing (**11**); a connection cylindrical member (**11**B) cylindrically extending backwardly from a middle part of the motor housing (**11**) to be formed integrally with the motor housing (**11**); a pair of second supports (**21**) integrally formed at right and left ends of the connection cylindrical member (**11**B); and a curved guide (**23**) having a front portion integrally formed with a lower end of the connection cylindrical member (**11**B) and integrally connecting upper parts of the pair of second supports (**21**), the curved guide (**23**) being configured to guide insertion of a rotor (**13**) into an insertion space (**11**B′) of the connection cylindrical member (**11**B), wherein a rear bearing cover (**19**) is coupled to an insertion space (**11**B′) in the connection cylindrical member (**11**B).
2. The motor of claim 1, wherein a support reinforcer (**24**) having a plurality of hollows is integrally formed downwardly of the curved guide (**23**).
3. The motor of claim 2, wherein a movement preventing support pad (**25**) being in surface-contact with a bottom surface of a main body of a washing machine is integrally formed in a lower part of the support reinforcer (**24**).
4. The motor of claim 3, wherein a nonslip embossed part (**25**A) is formed in a lower surface of the movement preventing support pad (**25**).